**Increment 3**

In increment 3 all validation checks on increment two are required, including invalid values for all the fields (such as amount, date, input choice etc.).

Increment 3 shall allow the loan seekers (borrowers) to enter their application details in multiple sessions; a user may leave the application submission process at a given time and may complete the application submission at a later stage. In case an existing application has to be completed the user is asked to provide the previous application id and the corresponding cnic for the application that is required to be completed/updated. Only the incomplete applications can be updated. Application marked as completed cannot be modified. You might have to add record incomplete status of an application in the file applications.txt. In order to achieve this functionality the application submission form may be divided into four categories:

- Personal Information
- Financial Information
- References
- Documents

These four sections may serve as checkpoints. When a user completes one section, the system saves the data in the applications.txt. If a user leaves, the system shall ask the user to enter the application id and the corresponding cnic; once this information is provided the system opens the incomplete application (from the previously saved checkpoint) and asks the user to complete the rest of form by entering the missing information. The fields in these sections are as follows:

Personal Information:

- Full Name
- Father's Name
- Postal Address
- Contact number
- Email address
- CNIC number (without dashes)
- CNIC expiry date (DD-MM-YYYY format)
- Employment status (Self-employed, salaried, retired, unemployed)
- Marital status (Single, Married, Divorced, Widowed)
- Gender
- Number of dependents

Financial Information

- Annual income in PKR (only numeric value without commas)
- Monthly average of electricity bill for the last 12 months (in PKR) (only numeric values without commas)
- The exact amount in PKR of the current electricity bill (only numeric values without commas)
- Information about any existing loans: If a user has taken any loans before the system shall record the following:

- Loan period is Active or not
- Total Loan Amount:
- The amount returned
- The loan amount still due
- Bank name
- Loan category (Car, Home, Bike)

References

- Details of references:
  - Name of Referee 1
  - CNIC of Referee 1
  - Issue date of CNIC of Referee 1
  - Phone number of Referee 1
  - Email address of Referee 1
  - Name of Referee 2
  - CNIC of Referee 2
  - Issue date of CNIC of Referee 2
  - Phone number of Referee 2
  - Email address of Referee 2

Documents

- Picture of applicant's CNIC Front
- Picture of applicant's CNIC Back
- Picture of applicant's recent electricity bill
- Picture of salary slip/Bank statement page 1

**Hint:** You may save C1, C2, C3 as application status in the applications.txt to record the application status. C3 shall be changed to Submitted once the application is complete.

Increment 3 requires a separate computer program (for lender) running in parallel to the bot. As mentioned in increment 2, this separate interface is a conventional interface and not a chat bot. The lender shall be able to change status of an existing application (i.e. approve or reject the applications).

In addition to the above the chatbot shall enter the general conversation mode (you may add an utterance in utterances,txt). In this mode the bot shall be able to have conversation with the users. To facilitate this human_chat_corpus.txt has been made available. Instead of separating the question and answers by a #, the interactions are written on separate lines. Human 1 means question, Human 2 is the response. Your chatbot shall match the user's input with the Human 1 text and shall provide a response which is the text from Human 2. The mechanism to retrieve the relevant answer is based on Intersection over Union (IoU). IoU is defined as follows:

Tokens are the terms in each sentence tokenized by space. Cardinality or the size of a set is indicated by enclosing a set in vertical lines. In simple language the numerator is the number of tokens common in two sets, whereas the denominator is the number of unique tokens in both sets.

Note the Human 1 line that has the highest value of IoU, its corresponding Human 2 line shall be selected as the chatbot's response.

**Notes:**

- Make a Personal.txt file similar to Home.txt, Utterances.txt, Car.txt, and Scooter.txt
- Make a sprint that ends on Wednesday November 26, 2025.
- Update requirements on Jira.
- All code should be updated on GitHub.
- There will be a demo on November 27, 2025 in regular class.
- As you are saving the information about your effort in each functionality/task. In the same google sheet, enter the estimated effort for each task before start working on a task. Then save the actual effort spent on the task. Notice and record the difference in the planned and actual effort. There are no penalties/incentives in having large or small differences. The marks are based on correctness of these data.